

NovaDiff — Baseline codebase

Baseline: 9aadd5eb5604

Target: NovaDiff

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Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI baseline codebase narrative

Repository overview

The repository is a monorepo that contains three main parts:

- **packages/graph-core/** – a TypeScript library that builds and normalises a graph of a codebase.
- **electron/** – the Electron desktop application (main, preload, renderer, and many helper modules).
- **cli/** – a Rust binary (`compare`) that performs diff analysis and is invoked from Electron.

The project is built with `npm / yarn` for the JavaScript parts and `cargo` for the Rust CLI.

Architectural layout



- **Electron main** (`electron/main.cjs`) orchestrates windows, menus, and IPC.
- **Preload** (`electron/preload.cjs`) exposes safe APIs to the renderer.
- **Renderer** (`index.html` + bundled JS) displays the UI.
- **Graph Core** is a pure TS library that builds a graph of files, fingerprints, and LLM-derived metadata.
- **CLI** (`cli/src/main.rs`) runs the compare engine and is called from Electron.

Key subsystems

Subsystem	Purpose	Main files	Notable symbols
Graph Core	Build and normalise a project graph	<code>packages/graph-core/src/analyzer/*.ts</code>	<code>GraphBuilder</code> , <code>normalizeNodeId</code> , <code>classifyUpdate</code> , <code>SemanticSearchEngine</code>
Electron Main	Application lifecycle, window management, progress reporting	<code>electron/main.cjs</code>	<code>createWindow</code> , <code>buildMenu</code> , <code>sendEngineProgress</code>
Electron Preload	Secure IPC bridge	<code>electron/preload.cjs</code>	<code>contextBridge</code> , <code>ipcRenderer</code>
Git Service	Git plumbing (status, blame, commit)	<code>electron/git-service.cjs</code>	<code>gitExecutable</code> , <code>runGit</code> , <code>parsePorcelainStatus</code>
GitHub Service	GitHub API interactions	<code>electron/github-service.cjs</code>	<code>runGh</code> , <code>listRepos</code> , <code>createPullRequest</code>

Subsystem	Purpose	Main files	Notable symbols
LLM	LLM request/response handling	<code>electron/llm.cjs</code>	<code>normalizeBase</code> , <code>sendAccumulated</code> , <code>ollamaGenerationOptions</code>
Knowledge Graph Runner	Executes graph core in a child process	<code>electron/knowledge-graph-runner.cjs</code>	<code>runCommand</code> , <code>loadCore</code> , <code>detectLanguage</code>
Compare Runner	Invokes the Rust compare engine	<code>electron/compare-runner.cjs</code>	<code>runCompareEngine</code> , <code>parseEngineStdout</code>
Export / Summary	Snapshot export, summary rendering	<code>electron/export-snapshot.cjs</code> , <code>electron/file-summary-export.cjs</code>	<code>exportProjectSnapshotZip</code> , <code>markdownToHtml</code>
CLI	Rust binary for compare engine	<code>cli/src/main.rs</code>	<code>ChangeKind</code> , <code>path_segments</code> , <code>subtree_change_counts</code>

Dependency signals

- **Cross-file call edges:** 228 edges indicate tight coupling between Electron modules and the graph core.
- **Resolved imports:** 43 edges, e.g.
 - `electron/knowledge-graph-runner.cjs` imports `packages/graph-core/src/analyzer/graph-builder.ts`.
 - `electron/compare-runner.cjs` calls the Rust binary via `resolveRustCli`.
 - `electron/git-service.cjs` imports `electron/git-blame.cjs`.
- **Symbol usage:**
 - `GraphBuilder` is instantiated in `electron/knowledge-graph-runner.cjs` and in tests under `packages/graph-core/src/__tests__`.
 - `classifyUpdate` is called by `electron/compare-runner.cjs`.
 - `SemanticSearchEngine` is used in `electron/llm.cjs`.

These signals show that the Electron UI consumes the graph core and that the CLI provides diff analysis.

Operational considerations

Task	Command / Script	Notes
Install dependencies	<code>npm install</code> (or <code>yarn</code>)	Installs TS, Electron, Rust toolchain, and LLM dependencies.
Build Electron app	<code>npm run build:electron</code>	Bundles main, preload, and renderer code.
Run Electron app	<code>npm start</code>	Launches the desktop UI.
Run CLI	<code>cargo run --bin compare</code>	Executes the compare engine; used by Electron.
Run tests	<code>npm test</code>	Executes Jest tests for graph core.
Generate alphabetical catalog	<code>node packages/graph-core/scripts/generate-alphabetical-catalog.mjs</code>	Produces a catalog of WASM modules.
Export snapshot	<code>electron/export-snapshot.cjs</code>	Creates a ZIP of the current project snapshot.
LLM integration	<code>electron/llm.cjs</code>	Requires an LLM backend (Ollama, LM Studio).
GitHub authentication	<code>electron/github-auth-flow.cjs</code>	Uses GitHub device flow.

- The Electron process spawns child processes for the graph core and the Rust CLI; ensure `node` and `cargo` are in `PATH`.
- LLM calls are optional; the app can operate with local analysis only.
- The `.novadiff-graph/` directory holds snapshot data; it is regenerated on each run.

Documentation gaps

- **CLI API** – The Rust binary's command-line options are not documented; only the `ChangeKind` enum is exposed.
- **LLM configuration** – No README section explains how to set up Ollama or LM Studio endpoints.
- **Electron IPC contracts** – The preload exposes many APIs, but the contract definitions are missing.

- **Graph Core public API** – The exported functions (`GraphBuilder` , `normalizeNodeId` , etc.) lack usage examples.
- **Testing strategy** – The test suite covers core logic but does not explain integration tests for Electron or CLI.
- **Build scripts** – `package.json` scripts are present but not described in the README.

Adding these details would improve onboarding and reduce friction for contributors.

Suggested onboarding and verification

1. Clone and install

```
git clone <repo>
cd <repo>
npm install
cargo build
```

2. Run the Electron app

```
npm start
```

Verify that the window opens and the menu bar appears.

3. Execute a sample compare

```
npm run compare -- --repo <path>
```

Check that the CLI outputs a diff summary.

4. Run tests

```
npm test
```

All Jest tests in `packages/graph-core/src/__tests__` should pass.

5. Generate catalog

```
node packages/graph-core/scripts/generate-alphabetical-catalog.mjs
```

Confirm that `alphabetical-catalog.json` is created.

6. Verify LLM flow (optional)

- Start an Ollama server (`ollama serve`).
- Set `LLM_ENDPOINT=localhost:11434` in `.env` .
- Run the Electron app and trigger a LLM-based analysis.

7. Inspect IPC

Open the renderer console (`Ctrl+Shift+I`) and check that IPC calls to `git-service` , `github-service` , and `llm` return expected data.

8. Review snapshot export

```
electron/export-snapshot.cjs
```

Ensure a ZIP file is produced and contains the expected files.

Baseline codebase metrics

Baseline codebase metrics

Repository: 9aadd5eb5604

Path: /Users/iamgroot/Library/Application Support/novadiff/novadiff-workspaces/7d1b19ef-9547-48e8-92d5-c8beabe3849c/snapshots/9aadd5eb5604

Metric	Value
Indexed files	359
Symbol-bearing files	80
Relative import edges	43
Cross-file call edges (heuristic)	228

Top extensions

Extension Count

.ts	210
.tsx	82
.cjs	27
.json	13
.css	5
.mjs	4
.svg	4
.md	3
(no ext)	2
.html	2
.ico	1
.jpg	1
.lock	1
.png	1
.rs	1
.toml	1
.txt	1

Top directories

Directory	Files
packages	235
src	71
electron	27
.	14
.novadiff-graph	4
cli	3
public	2
tests	2
scripts	1

Detected project stacks

- **node:** package.json

Sample files

- .gitignore
- .novadiff-graph/config.json
- .novadiff-graph/diff-overlay.json
- .novadiff-graph/knowledge-graph.json
- .novadiff-graph/meta.json
- CONTRIBUTIONS.md
- LICENSE
- README.md
- THIRD_PARTY_NOTICES.md
- cli/Cargo.lock
- cli/Cargo.toml
- cli/src/main.rs
- electron/code-city-model.cjs
- electron/compare-runner.cjs
- electron/diff-excerpt.cjs
- electron/export-snapshot.cjs
- electron/file-summary-export.cjs
- electron/gh-install.cjs
- electron/gh-path.cjs
- electron/git-blame.cjs
- electron/git-commit.cjs
- electron/git-publish.cjs
- electron/git-service.cjs
- electron/github-auth-flow.cjs
- electron/github-commit-context.cjs
- electron/github-service.cjs
- electron/graph/extract-structure.mjs
- electron/knowledge-graph-runner.cjs
- electron/llm.cjs
- electron/main.cjs
- electron/novadiff-docs-html.cjs
- electron/novadiff-docs-pdf.cjs
- electron/novadiff-docs-writer.cjs
- electron/prefetch-summaries.cjs
- electron/preload.cjs
- electron/repo-discovery.cjs
- electron/workspace-files.cjs
- electron/workspace-history.cjs
- electron/workspace-store.cjs
- index.html
- nova-diff-desktop.jpg
- nova-diff-icon.png
- package-lock.json
- package.json
- packages/graph-core/package.json
- packages/graph-core/scripts/generate-alphabetical-catalog.mjs
- packages/graph-core/scripts/language-names.txt
- packages/graph-core/src/__tests__/change-classifier.test.ts
- packages/graph-core/src/__tests__/domain-normalize.test.ts
- packages/graph-core/src/__tests__/domain-persistence.test.ts
- packages/graph-core/src/__tests__/domain-types.test.ts
- packages/graph-core/src/__tests__/embedding-search.test.ts
- packages/graph-core/src/__tests__/fingerprint.test.ts
- packages/graph-core/src/__tests__/framework-registry.test.ts
- packages/graph-core/src/__tests__/ignore-filter.test.ts
- packages/graph-core/src/__tests__/ignore-generator.test.ts
- packages/graph-core/src/__tests__/language-lesson.test.ts

- packages/graph-core/src/__tests__/language-registry.test.ts
- packages/graph-core/src/__tests__/layer-detector.test.ts
- packages/graph-core/src/__tests__/normalize-graph.test.ts
- packages/graph-core/src/__tests__/parsers.test.ts
- packages/graph-core/src/__tests__/plugin-discovery.test.ts
- packages/graph-core/src/__tests__/plugin-registry.test.ts
- packages/graph-core/src/__tests__/schema.test.ts
- packages/graph-core/src/__tests__/search.test.ts
- packages/graph-core/src/__tests__/staleness.test.ts
- packages/graph-core/src/__tests__/tour-generator.test.ts
- packages/graph-core/src/analyzer/graph-builder.test.ts
- packages/graph-core/src/analyzer/graph-builder.ts
- packages/graph-core/src/analyzer/language-lesson.ts
- packages/graph-core/src/analyzer/layer-detector.ts
- packages/graph-core/src/analyzer/llm-analyzer.test.ts
- packages/graph-core/src/analyzer/llm-analyzer.ts
- packages/graph-core/src/analyzer/normalize-graph.ts
- packages/graph-core/src/analyzer/tour-generator.ts
- packages/graph-core/src/change-classifier.ts
- packages/graph-core/src/embedding-search.ts
- packages/graph-core/src/fingerprint.ts
- packages/graph-core/src/ignore-filter.ts
- packages/graph-core/src/ignore-generator.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, path_segments, path_depth, parent_dir, subtree_change_counts, try_build_gitignore_for_root, is_novadiff_docs_reserved_rel, filter_out_novadiff_docs_changes, walk_path_allowed, filter_changes_for_prefetch_gitignore, is_utf8_text_file, is_change_text_eligible, filter_out_non_text_changes, prefetch_summary_queue, empty_summary_evidence, compact_snippet, push_unique_string, line_range_label, detect_file_roles, build_changed_line_ranges, changed_lines_for_side, collect_touched_symbols, is_import_like_line, is_export_like_line, collect_changed_decl_lines, collect_cited_changed_lines, build_verification_hints, build_evidence_badges
electron/code-city-model.cjs	normalizeSide, dominantAuthor, ownershipForRange, selectBlamePaths, buildBlameMap, buildSymbolFileIndex, buildSideNodes, buildCodeCityModelPayload
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine, parseEngineStdout, runCompareEngineAsync
electron/diff-excerpt.cjs	buildDiffExcerpt, buildDiffExcerptChunks
electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/file-summary-export.cjs	escapeHtml, escapeMermaidForHtml, injectMermaidFences, markdownToHtml, isReservedNovadiffRel, safeSummarySegments, resolveSummaryDirAbs, safeArtifactSegment, resolveSelectionBaseDirAbs, resolveSelectionDirAbs, buildFileSummaryDocument, buildSelectionSummaryDocument, exportFileSummaryArtifacts, readFileSummaryMarkdowns, exportSelectionSummaryArtifacts, readSelectionSummaryMarkdowns
electron/gh-install.cjs	resolveBrew, getInstallPlan, readGhVersion, getGhToolingStatus, runInstallCommand, installGh
electron/gh-path.cjs	homeDir, extraPathDirs, augmentPathForCli, ghCandidatePaths, resolveViaLoginShell, verifyGh, resolveGhExecutable, clearGhCache, isGhInstalled, ghInstallHint, ghNotFoundError
electron/git-blame.cjs	parseBlamePorcelain, blameFileAtRef, blameFileOwnership
electron/git-commit.cjs	getCommitDetail
electron/git-publish.cjs	buildFullCommitMessage, previewPublish, executePublish
electron/git-service.cjs	gitExecutable, runGit, tryRunGit, isGitRepo, parsePorcelainStatus, getRepoStatus, parseGithubSlugFromUrl, stageAll, stagePaths, unstagePaths, stageDistrict, commit, push, ensureWorktreeBase, removeDirSafe, checkoutRefWorktree, cleanupNovadiffWorktrees, preparePrCompareRoots, detectGitTooling
electron/github-auth-flow.cjs	stripAnsi, cancelGithubDeviceAuth, startGithubDeviceAuth
electron/github-commit-context.cjs	parseJsonSafe, shortGhError, normalizeThread, getGithubCommitContext
electron/github-service.cjs	runGh, tryRunGh, isGhAvailable, getAuthStatus, listRepos, listPullRequests, viewPullRequest, createPullRequest, getUserProfile, listDetectedAccounts, slugFromRepoRoot
electron/graph/extract-structure.mjs	main, buildResult
electron/knowledge-graph-runner.cjs	resolveRepoRoot, resolveNodeExecutable, nodeChildEnv, runCommand, ensureGraphCoreBuilt, loadCore, isTextCandidate, walkProjectFiles, detectLanguage, fileCategory, resolveRelativeImport, buildBatchImportData, runExtractStructureBatch, mapStructuralAnalysis, heuristicComplexity, buildFileMeta, topSegment, buildLayersFromFiles, assignNodeLayers, buildDiffOverlay, getGitHash, buildKnowledgeGraph
electron/llm.cjs	normalizeBase, sendAccumulated, normalizeRepeatUnit, arraysEqual, trimRepeatedParagraphSuffix, trimRepeatedLineSuffix, sanitizeModelResponse, ollamaGenerationOptions, lmStudioGenerationOptions, ollamaNumPredict, streamOllamaChat, lmStudioMaxTokens, streamLmStudioChat, buildWorkspaceDocPrompt, buildCodebaseDocPrompt, buildCommitMessagePrompt, buildSelectionDocPrompt, bulletBlock, renderSummaryEvidenceBlock, buildExplainCodePrompt, buildGroundingContext, verifierWordLimit, buildGroundedRevisionPrompt, shouldRunVerifier, reviseGroundedMarkdown, buildPerFileSummarizePrompt, buildChunkExtractionPrompt, buildChunkSynthesisPrompt

File	Symbols / defs (regex)
electron/main.cjs	sendEngineProgress, getTargetWindow, windowStatePayload, emitWindowState, iconPath, buildMenu, createWindow, sendKnowledgeGraphProgress, sendWorkspaceHistoryProgress, sendGithubAuthProgress
electron/novadiff-docs-html.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, escapeRegex, countMatches, normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, injectMermaidFences, normalizeMarkdownDocument, stripTags, slugify, addHeadingAnchors, markdownToHtml, mermaidSection, navHtml, topDirectory, buildImpactHeatmapHtml, buildSearchIndex, buildEnhancementScript, wrapPage, bundleMetaLine, buildIndexHtml, buildNarrativeHtml, buildMetricsHtml
electron/novadiff-docs-pdf.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, renderSymbolsTable, renderImportEdgesTable, renderDetectedProjects, renderCrossFileCallEdges, mermaidBlock, buildMainPdfHtml, buildDiagramsPdfHtml, waitForPdfReady, htmlToPdfFile, generateNovadiffDocsPdf
electron/novadiff-docs-writer.cjs	bundleKeyOf, bundleBaseDir, bundleHeadline, readme, writeNovadiffDocsBundle
electron/prefetch-summaries.cjs	clearSummaryPrefetchCache, stopSummaryPrefetchWorker, getPrefetchedSummary, defaultPrefetchLimit, runSummaryPrefetchLoop, startSummaryPrefetchWorker
electron/repo-discovery.cjs	homeSearchRoots, readGitRemoteSlug, scanForGitRepos, discoverRepos, matchRepoForGithubRepo
electron/workspace-files.cjs	listSnapshotFiles, readSnapshotTextFile
electron/workspace-history.cjs	removeDirSafe, isUsableSnapshotDir, listCommitsOldestFirst, snapshotCommit, ensureCommitSnapshot, indexWorkspaceHistory, refreshWorkspaceHistory
electron/workspace-store.cjs	sessionPath, workspacesRoot, readJson, withSessionWriteLock, writeJsonAtomic, writeJson, loadSession, setLocalOnlyMode, saveSession, listWorkspaces, upsertWorkspace, setActiveWorkspace, setGitUser, getWorkspace, updateWorkspaceUiState, updateWorkspaceLiveRepo, createWorkspace
packages/graph-core/scripts/generate-alphabetical-catalog.mjs	normalizeKey, slugify, resolveWasmsKey, resolveExtensions, claimExtensions
packages/graph-core/src/__tests__/change-classifier.test.ts	makeAnalysis
packages/graph-core/src/__tests__/plugin-registry.test.ts	createMockPlugin
packages/graph-core/src/__tests__/staleness.test.ts	makeGraph
packages/graph-core/src/analyzer/graph-builder.ts	FileMeta, FileAnalysisMeta, NonCodeFileMeta, NonCodeFileAnalysisMeta, GraphBuilder
packages/graph-core/src/analyzer/language-lesson.ts	LanguageLessonResult, buildConceptPatterns, detectLanguageConcepts, getLanguageDisplayName, buildLanguageLessonPrompt, extractJson, parseLanguageLessonResponse
packages/graph-core/src/analyzer/layer-detector.ts	LLMLayerResponse, toLayerId, matchFileToLayer, detectLayers, buildLayerDetectionPrompt, parseLayerDetectionResponse, applyLLMLayers
packages/graph-core/src/analyzer/llm-analyzer.ts	LLMFileAnalysis, LLMProjectSummary, buildFileAnalysisPrompt, buildProjectSummaryPrompt, extractJson, parseFileAnalysisResponse, parseProjectSummaryResponse
packages/graph-core/src/analyzer/normalize-graph.ts	stripToValidPrefix, normalizeNodeId, normalizeComplexity, DroppedEdge, NormalizationStats, NormalizeBatchResult, inferTypeFromId, normalizeBatchOutput
packages/graph-core/src/analyzer/tour-generator.ts	buildTourGenerationPrompt, parseTourGenerationResponse, generateHeuristicTour
packages/graph-core/src/change-classifier.ts	UpdateDecision, classifyUpdate, detectDirectoryChanges, topDirectory, summarizeChanges
packages/graph-core/src/embedding-search.ts	SemanticSearchOptions, cosineSimilarity, SemanticSearchEngine
packages/graph-core/src/fingerprint.ts	FunctionFingerprint, ClassFingerprint, ImportFingerprint, FileFingerprint, FingerprintStore, FileChangeResult, ChangeAnalysis, contentHash, extractFileFingerprint, compareFingerprints, buildFingerprintStore, analyzeChanges
packages/graph-core/src/ignore-filter.ts	IgnoreFilter, createIgnoreFilter

File	Symbols / defs (regex)
packages/graph-core/src/ignore-generator.ts	parseGitIgnorePatterns, isCoveredByDefaults, generateStarterIgnoreFile
packages/graph-core/src/languages/catalog-builder.ts	catalogEntryToLanguageConfig, buildAlphabeticalLanguageConfigs, listAlphabeticalLanguages
packages/graph-core/src/languages/catalog-types.ts	LanguageCatalogEntry
packages/graph-core/src/languages/framework-registry.ts	FrameworkRegistry
packages/graph-core/src/languages/language-registry.ts	LanguageRegistry
packages/graph-core/src/languages/merge-configs.ts	mergeLanguageConfigs, languageConfigsForRegistry
packages/graph-core/src/persistence/index.ts	ensureDir, sanitiseFilePaths, saveGraph, loadGraph, saveMeta, loadMeta, saveFingerprints, loadFingerprints, saveConfig, loadConfig, saveDomainGraph, loadDomainGraph
packages/graph-core/src/plugins/discovery.ts	PluginEntry, PluginConfig, parsePluginConfig, serializePluginConfig
packages/graph-core/src/plugins/extractors/__tests__/cpp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/csharp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/go-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/java-extractor.test.ts	parse, Repository
packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	parse, helper, greet, noReturn, noop, multiline, Loggable, Repository, publicFunc, anotherFunc, myFunc, process, main, helper, findUser
packages/graph-core/src/plugins/extractors/__tests__/python-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/ruby-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/rust-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/base-extractor.ts	traverse, getStringValue, findChild, findChildren, hasChildOfType
packages/graph-core/src/plugins/extractors/cpp-extractor.ts	unwrapDeclaratorName, extractFuncDeclName, extractParams, extractReturnType, isStatic, CppExtractor
packages/graph-core/src/plugins/extractors/csharp-extractor.ts	extractParams, extractReturnType, hasModifier, extractUsingSource, lastComponent, extractInvocationName, CSharpExtractor
packages/graph-core/src/plugins/extractors/generic-extractor.ts	nodeName, lineRange, importSource, GenericExtractor
packages/graph-core/src/plugins/extractors/go-extractor.ts	extractParams, extractResultType, extractReceiverType, isExported, GoExtractor
packages/graph-core/src/plugins/extractors/java-extractor.ts	extractParams, extractReturnType, hasModifier, extractScopedIdentifierPath, lastComponent, JavaExtractor
packages/graph-core/src/plugins/extractors/php-extractor.ts	extractParams, extractReturnType, extractUseName, lastSegment, PhpExtractor
packages/graph-core/src/plugins/extractors/python-extractor.ts	extractParams, extractReturnType, unwrapDecorated, PythonExtractor
packages/graph-core/src/plugins/extractors/ruby-extractor.ts	extractParams, extractAttrProperties, getStringContent, RubyExtractor
packages/graph-core/src/plugins/extractors/rust-extractor.ts	extractParams, extractReturnType, isPublic, extractScopedPath, RustExtractor

File	Symbols / defs (regex)
packages/graph-core/src/plugins/extractors/types.ts	LanguageExtractor
packages/graph-core/src/plugins/extractors/typescript-extractor.ts	extractParams, extractReturnType, extractImportSpecifiers, TypeScriptExtractor
packages/graph-core/src/plugins/parsers/dockerfile-parser.ts	DockerfileParser

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a "call graph"-style overview at module boundaries.

From	To
electron/code-city-model.cjs	electron/git-blame.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs
electron/gh-install.cjs	electron/gh-path.cjs
electron/git-commit.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/github-service.cjs
electron/github-auth-flow.cjs	electron/gh-path.cjs
electron/github-commit-context.cjs	electron/github-service.cjs
electron/github-service.cjs	electron/git-service.cjs
electron/github-service.cjs	electron/gh-path.cjs
electron/github-service.cjs	electron/github-commit-context.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
electron/main.cjs	electron/prefetch-summaries.cjs
electron/main.cjs	electron/novadiff-docs-writer.cjs
electron/main.cjs	electron/novadiff-docs-pdf.cjs
electron/main.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/file-summary-export.cjs
electron/main.cjs	electron/code-city-model.cjs
electron/main.cjs	electron/knowledge-graph-runner.cjs
electron/main.cjs	electron/git-service.cjs
electron/main.cjs	electron/github-service.cjs
electron/main.cjs	electron/git-commit.cjs
electron/main.cjs	electron/repo-discovery.cjs
electron/main.cjs	electron/git-publish.cjs
electron/main.cjs	electron/workspace-store.cjs
electron/main.cjs	electron/git-blame.cjs
electron/main.cjs	electron/workspace-files.cjs
electron/main.cjs	electron/workspace-history.cjs
electron/main.cjs	electron/github-auth-flow.cjs
electron/main.cjs	electron/gh-path.cjs
electron/main.cjs	electron/gh-install.cjs
electron/main.cjs	electron/export-snapshot.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/prefetch-summaries.cjs	electron/llm.cjs
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs
electron/prefetch-summaries.cjs	electron/git-service.cjs
electron/repo-discovery.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
packages/graph-core/src/__tests__/ignore-filter.test.ts	packages/graph-core/src/ignore-filter.ts
packages/graph-core/src/__tests__/ignore-generator.test.ts	packages/graph-core/src/ignore-generator.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
cli/src/main.rs	electron/git-service.cjs	push
electron/code-city-model.cjs	electron/git-blame.cjs	blameFileOwnership
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs	normalizeMarkdownDocument
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs	htmlToPdfFile
electron/gh-install.cjs	electron/gh-path.cjs	augmentPathForCli
electron/gh-install.cjs	electron/gh-path.cjs	isGhInstalled
electron/gh-install.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/gh-install.cjs	electron/gh-path.cjs	clearGhCache
electron/git-commit.cjs	electron/git-service.cjs	runGit
electron/git-commit.cjs	electron/git-service.cjs	tryRunGit
electron/git-publish.cjs	electron/git-service.cjs	isGitRepo
electron/git-publish.cjs	electron/git-service.cjs	getRepoStatus
electron/git-publish.cjs	electron/github-service.cjs	getAuthStatus
electron/git-publish.cjs	electron/github-service.cjs	slugFromRepoRoot
electron/git-publish.cjs	electron/git-service.cjs	stagePaths
electron/git-publish.cjs	electron/git-service.cjs	stageAll
electron/git-publish.cjs	electron/git-service.cjs	commit
electron/git-publish.cjs	electron/git-service.cjs	push
electron/git-publish.cjs	electron/github-service.cjs	createPullRequest
electron/github-auth-flow.cjs	electron/gh-path.cjs	isGhInstalled
electron/github-auth-flow.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-auth-flow.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-auth-flow.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-commit-context.cjs	electron/github-service.cjs	getAuthStatus
electron/github-commit-context.cjs	electron/github-service.cjs	tryRunGh
electron/github-service.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-service.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-service.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-service.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/graph/extract-structure.mjs	packages/graph-core/src/plugins/parsers/index.ts	registerAllParsers
electron/knowledge-graph-runner.cjs	packages/graph-core/src/analyzer/graph-builder.ts	GraphBuilder
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveGraph
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveMeta
electron/llm.cjs	packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	process
electron/main.cjs	electron/gh-path.cjs	augmentPathForCli
electron/main.cjs	electron/compare-runner.cjs	runCompareEngineAsync

From file	To file	Callee
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider
electron/main.cjs	electron/prefetch-summaries.cjs	stopSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	clearSummaryPrefetchCache
electron/main.cjs	electron/prefetch-summaries.cjs	startSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	getPrefetchedSummary
electron/main.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readFileSummaryMarkdowns
electron/main.cjs	electron/file-summary-export.cjs	exportSelectionSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
electron/main.cjs	electron/code-city-model.cjs	buildCodeCityModelPayload
electron/main.cjs	electron/novadiff-docs-writer.cjs	writeNovadiffDocsBundle
electron/main.cjs	electron/novadiff-docs-html.cjs	writeNovadiffDocsHtml
electron/main.cjs	electron/novadiff-docs-pdf.cjs	generateNovadiffDocsPdf
electron/main.cjs	electron/novadiff-docs-html.cjs	refreshNovadiffDocsHtmlFromDisk
electron/main.cjs	electron/knowledge-graph-runner.cjs	buildKnowledgeGraph
electron/main.cjs	electron/git-service.cjs	detectGitTooling
electron/main.cjs	electron/github-service.cjs	getAuthStatus
electron/main.cjs	electron/git-service.cjs	getRepoStatus
electron/main.cjs	electron/repo-discovery.cjs	discoverRepos
electron/main.cjs	electron/repo-discovery.cjs	matchRepoForGithubRepo
electron/main.cjs	electron/github-service.cjs	listRepos
electron/main.cjs	electron/github-service.cjs	listPullRequests
electron/main.cjs	electron/git-service.cjs	isGitRepo
electron/main.cjs	electron/git-service.cjs	preparePrCompareRoots
electron/main.cjs	electron/github-service.cjs	viewPullRequest
electron/main.cjs	electron/git-publish.cjs	previewPublish
electron/main.cjs	electron/git-publish.cjs	executePublish
electron/main.cjs	electron/git-service.cjs	stagePaths
electron/main.cjs	electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/main.cjs	electron/git-service.cjs	stageDistrict
electron/main.cjs	electron/workspace-store.cjs	loadSession
electron/main.cjs	electron/github-service.cjs	listDetectedAccounts
electron/main.cjs	electron/github-service.cjs	getUserProfile
electron/main.cjs	electron/gh-install.cjs	getGhToolingStatus
electron/main.cjs	electron/gh-install.cjs	installGh
electron/main.cjs	electron/workspace-store.cjs	setLocalOnlyMode
electron/main.cjs	electron/github-auth-flow.cjs	startGithubDeviceAuth
electron/main.cjs	electron/github-auth-flow.cjs	cancelGithubDeviceAuth
electron/main.cjs	electron/workspace-store.cjs	setGitUser
electron/main.cjs	electron/workspace-store.cjs	workspacesRoot
electron/main.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/main.cjs	electron/workspace-store.cjs	createWorkspace
electron/main.cjs	electron/workspace-history.cjs	indexWorkspaceHistory
electron/main.cjs	electron/workspace-store.cjs	getWorkspace
electron/main.cjs	electron/workspace-store.cjs	setActiveWorkspace
electron/main.cjs	electron/workspace-store.cjs	listWorkspaces
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceLiveRepo
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceUiState
electron/main.cjs	electron/git-blame.cjs	blameFileAtRef
electron/main.cjs	electron/git-commit.cjs	getCommitDetail
electron/main.cjs	electron/github-commit-context.cjs	getGithubCommitContext

Codebase outline (JSON excerpt)

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    {
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    },
    {
      "count": 13,
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    {
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      "ext": ".css"
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    {
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      "ext": ".mjs"
    },
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    {
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  ],
  "class_diagram_mermaid": "classDiagram\n  class CSharpExtractor\n  class ChangeAnalysis\n  class ChangeKind\n  class ClassFingerprint\n  class CppExtractor\n  class DockerfileParser\n  class DroppedEdge\n  class EnvParser\n  class FileAnalysisMeta\n  class FileChangeResult\n  class FileFingerprint\n  class FileMeta\n  class FingerprintStore\n  class FrameworkRegistry\n  class FunctionFingerprint\n  class GenericExtractor\n  class GoExtractor\n  class GraphBuilder\n  class GraphQLParser\n  class IgnoreFilter\n  class ImportFingerprint\n  class JSONConfigParser\n  class
```



```

JavaExtractor\n  class LLMFileAnalysis",
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      "to_file": "electron/git-blame.cjs",
      "via": "blameFileOwnership"
    },
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      "to_file": "electron/novadiff-docs-html.cjs",
      "via": "normalizeMarkdownDocument"
    },
    {
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      "via": "htmlToPdfFile"
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      "to_file": "electron/gh-path.cjs",
      "via": "augmentPathForCli"
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      "to_file": "electron/gh-path.cjs",
      "via": "isGhInstalled"
    },
    {
      "from_file": "electron/gh-install.cjs",
      "to_file": "electron/gh-path.cjs",
      "via": "resolveGhExecutable"
    },
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      "via": "clearGhCache"
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      "to_file": "electron/git-service.cjs",
      "via": "stagePaths"
    },
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      "to_file": "electron/git-service.cjs",
      "via": "stageAll"
    }
  ],

```

```

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  "via": "commit"
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{
  "from_file": "electron/git-publish.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "push"
},
{
  "from_file": "electron/git-publish.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "createPullRequest"
},
{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "isGhInstalled"
},
{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "ghNotFoundError"
},
{
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  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
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  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
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  "to_file": "electron/github-service.cjs",
  "via": "tryRunGh"
},
{
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  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
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  "via": "parseGithubSlugFromUrl"
},
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  "to_file": "packages/graph-core/src/plugins/parsers/index.ts",
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},
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  "to_file": "packages/graph-core/src/analyzer/graph-builder.ts",
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},
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  "to_file": "packages/graph-core/src/persistence/index.ts",
  "via": "saveGraph"
},
{
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```

```

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  },
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    "to_file": "packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts",
    "via": "process"
  },
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    "to_file": "electron/gh-path.cjs",
    "via": "augmentPathForCli"
  },
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    "to_file": "electron/compare-runner.cjs",
    "via": "runCompareEngineAsync"
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  {
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    "to_file": "electron/compare-runner.cjs",
    "via": "runCompareEngine"
  },
  {
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    "to_file": "electron/llm.cjs",
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  },
  {
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  {
    "from_file": "electron/main.cjs",
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    "via": "startSummaryPrefetchWorker"
  },
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  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/file-summary-export.cjs",
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  },
  {
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    "to_file": "electron/file-summary-export.cjs",
    "via": "readFileSummaryMarkdowns"
  },
  {
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    "to_file": "electron/file-summary-export.cjs",
    "via": "exportSelectionSummaryArtifacts"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/file-summary-export.cjs",
    "via": "readSelectionSummaryMarkdowns"
  },
  {
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    "via": "buildCodeCityModelPayload"
  }

```

```

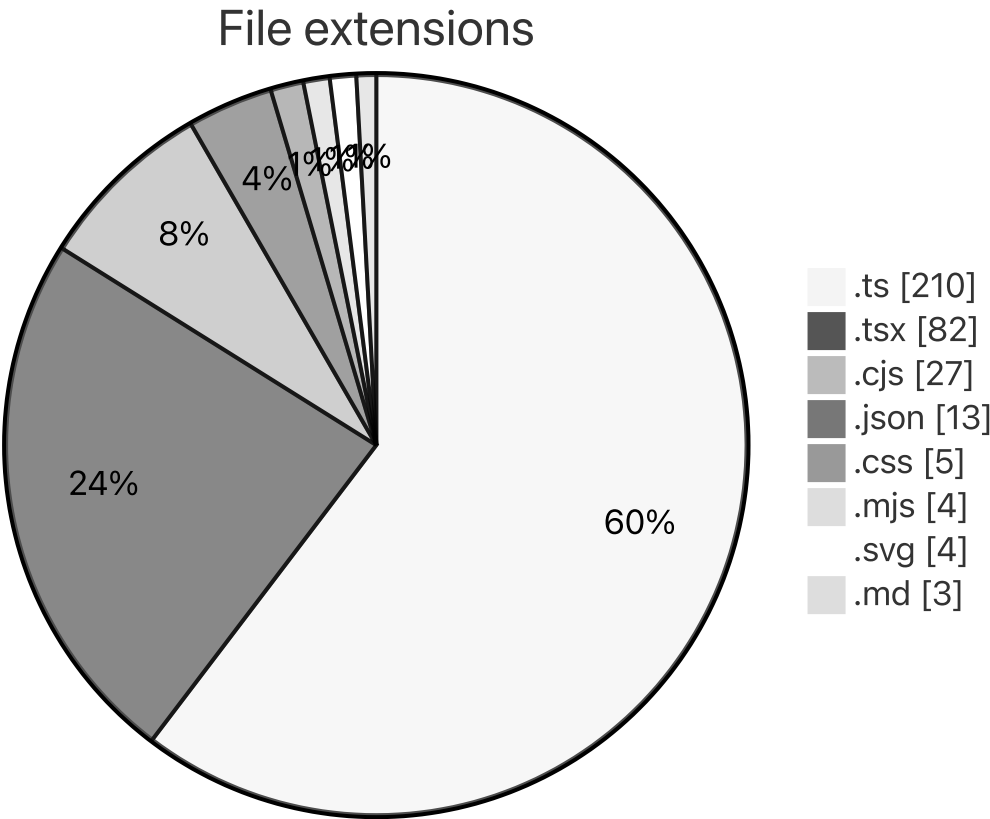
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},
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  "via": "listRepos"
},
{
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  "via": "listPullRequests"
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{
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  "via": "isGitRepo"
},
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  "to_file": "electron/git-service.cjs",
  "via": "preparePrCompareRoots"
},
{
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  "to_file": "electron/github-service.cjs",
  "via": "viewPullRequest"
},
{
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},
{

```

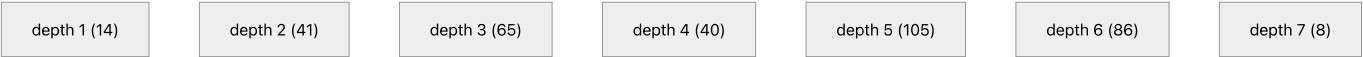
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    "via": "stageDistrict"
  },
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    "to_file": "electron/workspace-store.cjs",
    "via": "loadSession"
  },
  {
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    "to_file": "electron/github-service.cjs",
    "via": "listDetectedAccounts"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "getUserProfile"
  },
  {
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    "to_file": "electron/gh-install.cjs",
    "via": "getGhToolingStatus"
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  {
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    "to_file": "electron/gh-install.cjs",
    "via": "installGh"
  },
  {
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    "to_file": "electron/workspace-store.cjs",
    "via": "setLocalOnlyMode"
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Diagrams (Mermaid)

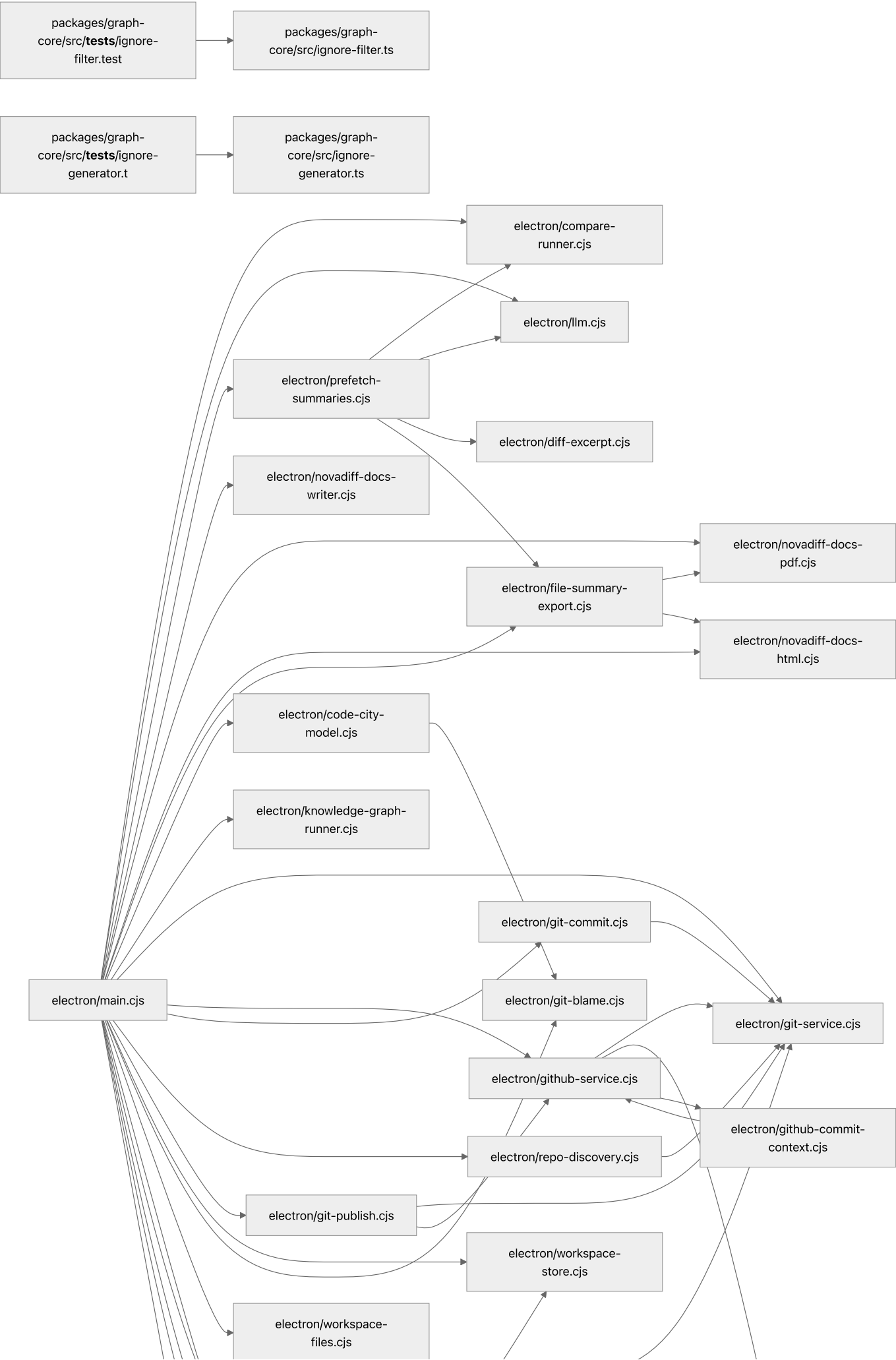
File extension mix

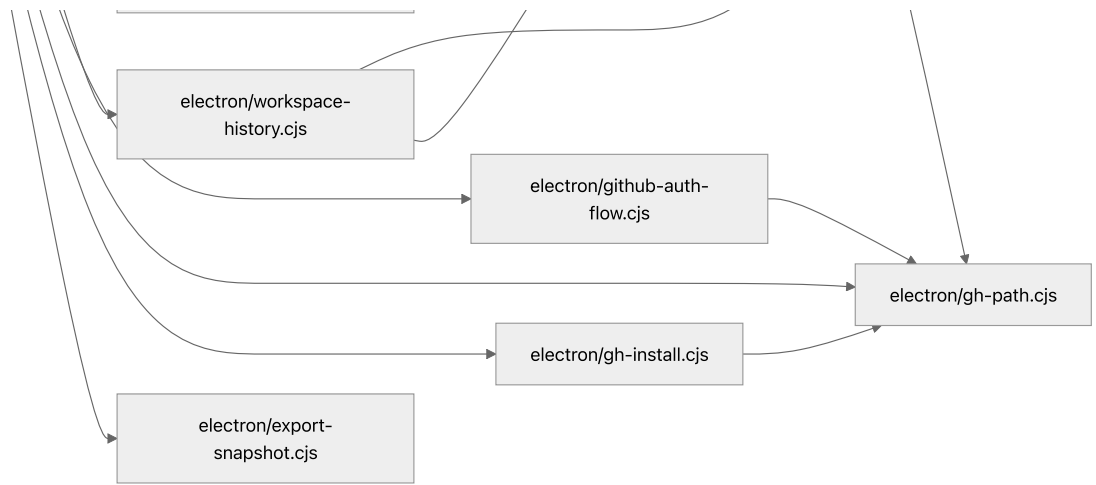


Directory depth



Module import graph (relative imports)





Cross-file calls (heuristic)

